Name : Arunima Sanjeev
Reg No : 23MIS0048

Lab Cat – exp 9,11

Experiment 11:

```
pipeline {
    agent any
    stages {

        stage('Version 1') {
            steps {
                echo '--- Running Version 1 Tasks ---'

                git branch: 'main', url: 'https://github.com/ArunimaSanjeev/LibraryManagement.git'


                echo 'Pipeline execution has started.'


                echo 'Pipeline execution has completed.'
            }
        }


        stage('Version 2') {
            steps {
                echo '--- Running Version 2 Tasks ---'

                git branch: 'main', url: 'https://github.com/ArunimaSanjeev/LibraryManagement.git'

                echo 'Project files are being processed...'

                echo 'Processing completion achieved.'
            }
        }

        stage('Version 3') {
            steps {
                echo '--- Running Version 3 Tasks ---'

                git branch: 'main', url: 'https://github.com/ArunimaSanjeev/LibraryManagement.git'


                bat 'echo Build Successful > status.txt'
                bat 'type status.txt'
            }
        }


        stage('Version 4') {
            steps {
                echo '--- Running Version 4 Tasks ---'

                git branch: 'main', url: 'https://github.com/ArunimaSanjeev/LibraryManagement.git'


                bat 'echo Pipeline Execution Completed >> status.txt'

                bat 'type status.txt'
            }
        }
```

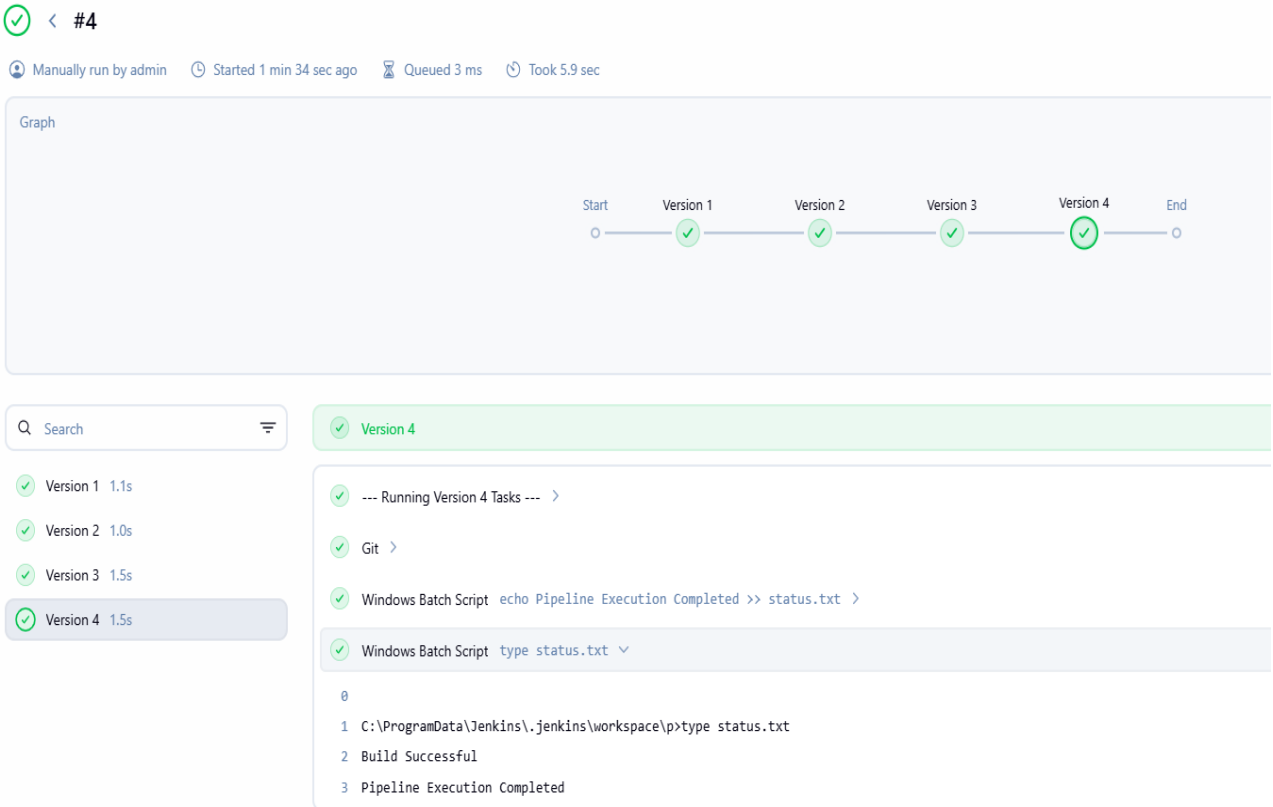
```
    }
}
```

```
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Version 4)
[Pipeline] echo
--- Running Version 4 Tasks ---
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\p\.git # timeout=10
Fetching changes from the remote Git repository
 > git.exe config remote.origin.url https://github.com/ArunimaSanjeev/LibraryManagement.git # timeout=10
Fetching upstream changes from https://github.com/ArunimaSanjeev/LibraryManagement.git
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/ArunimaSanjeev/LibraryManagement.git +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision de58a90335296ac860fb84eca4def68bc52c2f2b (refs/remotes/origin/main)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f de58a90335296ac860fb84eca4def68bc52c2f2b # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe branch -D main # timeout=10
 > git.exe checkout -b main de58a90335296ac860fb84eca4def68bc52c2f2b # timeout=10
Commit message: "new commit"
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\p>echo Pipeline Execution Completed  1>>status.txt
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\p>type status.txt
Build Successful
Pipeline Execution Completed
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

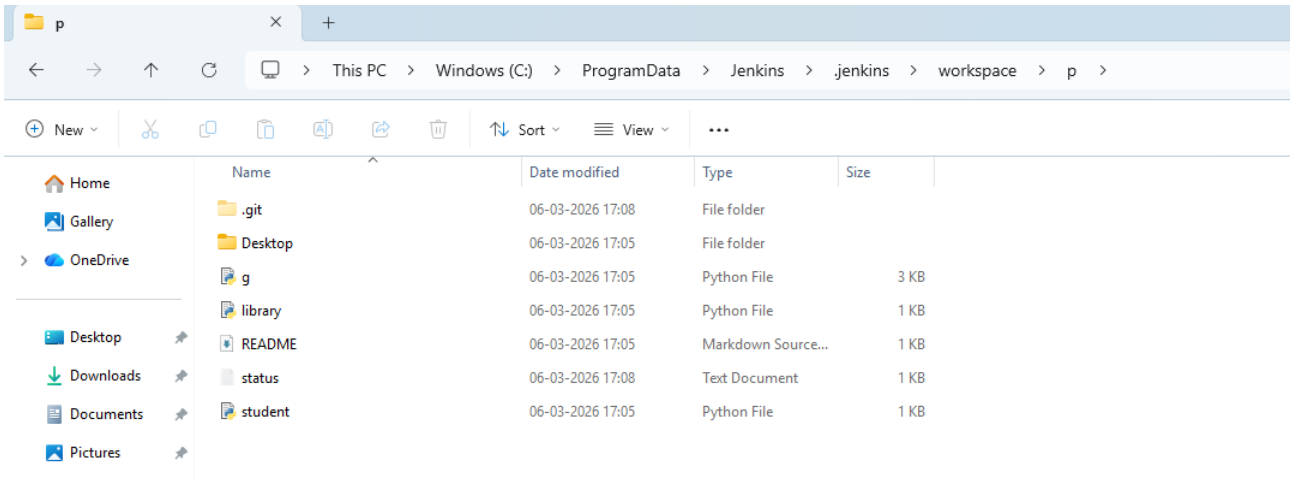
p

Stage View

	Version 1	Version 2	Version 3	Version 4
Average stage times: (full run time: ~5s)	1s	1s	1s	1s
#4 Mar 06 17:08 No Changes	1s	1s	1s	1s



workspace



Experiment 9:

```
pipeline {
   agent any
   stages {

      stage('Version 1') {
         steps {
            echo '--- Version 1: Directory Creation ---'

            git branch: 'main', url: 'https://github.com/ArunimaSanjeev/LibraryManagement.git'


            bat 'if not exist testfolder mkdir testfolder'


            bat 'dir'
```

```
        }
    }


    stage('Version 2') {
        steps {
            echo '--- Version 2: File Creation ---'

            git branch: 'main', url: 'https://github.com/ArunimaSanjeev/LibraryManagement.git'


            bat '''
            if not exist testfolder mkdir testfolder
            echo File 1 Content > testfolder\\file1.txt
            echo File 2 Content > testfolder\\file2.txt
            '''

            bat 'dir testfolder'
        }
    }


    stage('Version 3') {
        steps {
            echo '--- Version 3: Copy Operations ---'

            git branch: 'main', url: 'https://github.com/ArunimaSanjeev/LibraryManagement.git'


            bat 'copy README.md testfolder\\'

            bat 'dir testfolder'
        }
    }


    stage('Version 4') {
        steps {
            echo '--- Version 4: Directory Deletion ---'

            git branch: 'main', url: 'https://github.com/ArunimaSanjeev/LibraryManagement.git'


            bat 'if exist testfolder rd /s /q testfolder'


            bat 'dir'
        }
    }
  }
}
```

p2

Stage View

	Version 1	Version 2	Version 3	Version 4
Average stage times: (full run time: ~7s)	2s	1s	1s	1s
#1 Mar 06 17:13 No Changes	2s	1s	1s	1s

#1

Manually run by admin · Started 3 min 33 sec ago · Queued 4 ms · Took 7.6 sec

Rerun

Graph

Start → Version 1 → Version 2 → Version 3 → Version 4 → End

Search

- Version 1 2.2s
- Version 2 1.6s
- Version 3 1.5s
- Version 4 1.5s

Version 4 · 1.5s · Started 3m 27s ago · Jenkins

--- Version 4: Directory Deletion ---	7ms
Git	0.98s
Windows Batch Script `if exist testfolder rd /s /q testfolder`	0.27s
Windows Batch Script `dir`	0.27s

```
0
1 C:\ProgramData\Jenkins\.jenkins\workspace\p2>dir
2  Volume in drive C is Windows
3  Volume Serial Number is 368D-E1FD
4
5  Directory of C:\ProgramData\Jenkins\.jenkins\workspace\p2
6
7 06-03-2026  17:13    <DIR>          .
```

workspace:

p2

This PC > Windows (C:) > ProgramData > Jenkins > .jenkins > workspace > p2 >

New · Sort · View

- Home
- Gallery
- OneDrive
- Desktop
- Downloads
- Documents
- Pictures

Name	Date modified	Type	Size
.git	06-03-2026 17:13	File folder	
Desktop	06-03-2026 17:13	File folder	
g	06-03-2026 17:13	Python File	3 KB
library	06-03-2026 17:13	Python File	1 KB
README	06-03-2026 17:13	Markdown Source...	1 KB
student	06-03-2026 17:13	Python File	1 KB

Search

- Version 1 2.2s
- Version 2 1.6s
- Version 3 1.5s
- Version 4 1.5s

Version 4

--- Version 4: Directory Deletion ---

Git

Windows Batch Script `if exist testfolder rd /s /q testfolder`

Windows Batch Script `dir`

```
0
1 C:\ProgramData\Jenkins\.jenkins\workspace\p2>dir
2  Volume in drive C is Windows
3  Volume Serial Number is 368D-E1FD
4
5  Directory of C:\ProgramData\Jenkins\.jenkins\workspace\p2
6
7 06-03-2026  17:13    <DIR>          .
8 06-03-2026  17:13    <DIR>          ..
9 06-03-2026  17:13    <DIR>          Desktop
10 06-03-2026  17:13             2,341 g.py
11 06-03-2026  17:13               312 library.py
12 06-03-2026  17:13                28 README.md
13 06-03-2026  17:13                38 student.py
14                4 File(s)          2,719 bytes
15                3 Dir(s)  356,335,521,792 bytes free
```

```
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/ArunimaSanjeev/LibraryManagement.git +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision de58a90335296ac860fb84eca4def68bc52c2f2b (refs/remotes/origin/main)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f de58a90335296ac860fb84eca4def68bc52c2f2b # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe branch -D main # timeout=10
 > git.exe checkout -b main de58a90335296ac860fb84eca4def68bc52c2f2b # timeout=10
Commit message: "new commit"
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\p2>if exist testfolder rd /s /q testfolder
[Pipeline] bat (hide)

C:\ProgramData\Jenkins\.jenkins\workspace\p2>dir
 Volume in drive C is Windows
 Volume Serial Number is 368D-E1FD

 Directory of C:\ProgramData\Jenkins\.jenkins\workspace\p2

06-03-2026  17:13    <DIR>          .
06-03-2026  17:13    <DIR>          ..
06-03-2026  17:13    <DIR>          Desktop
06-03-2026  17:13             2,341 g.py
06-03-2026  17:13               312 library.py
06-03-2026  17:13                28 README.md
06-03-2026  17:13                38 student.py
               4 File(s)          2,719 bytes
               3 Dir(s)  356,335,521,792 bytes free
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```